ACHRAF RAHIBE

UAV & AVIONICS SYSTEMS ENGINEER

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UAV and avionics engineer developing complete aerial, ground and maritime vehicles: design, custom electronics, software integration and validation through autonomous missions. Leads multidisciplinary teams and client engineering work.

TECHNICAL SKILLS

Flight Control & Avionics
Pixhawk-family/H7; ArduPilot, PX4, INAV, Betaflight; Mission Planner, QGroundControl; PID/FFT.

Electronics & Prototyping
Altium Designer, KiCad: flight-controller, ESC and power PCBs; SMD/through-hole soldering, board testing and vehicle integration.

Embedded & Interfaces
C/C++, Python, Linux, ROS2, Git; STM32, CM5, Jetson Orin Nano; CAN/DroneCAN, UART, SPI, I2C.

Vision & Navigation
YOLOv8/ByteTrack; A8C gimbal, GNSS/RTK, GPS-denied UAVs, AIS/ADS-B.

Design & Simulation
Fusion 360, SolidWorks, Onshape; FDM, foam, CNC-cut wood; fibreglass/resin; carbon-fibre knowledge; custom-model Gazebo/SITL, HIL.

Design & Communication
Photoshop, Illustrator; graphic design and engineering consultations.

EDUCATION

State Engineer Diploma
Electrical Engineering and Telecommunications; specialisation in Electrical Engineering and Industrial Control. FSTM, Hassan II University.
2021-2025 | Bac+5

DUT in Electrical Engineering
Electronics and Industrial Computing. EST Fès, Université Sidi Mohamed Ben Abdellah.
2019-2021

Technical Baccalaureate
Electrical Sciences and Technology. Lycée Technique Anoual, Casablanca.
2018-2019

LEADERSHIP & TEACHING

ROBOTHINGS: Founder/President (2023-2025).
ENACTUS FSTM: Marketing & Graphic Design Lead (2022-2023).

Robotics instructor: centres and schools, including G.S. Kepler School.

LANGUAGES

Arabic: Native
English & French: Fluent
German: Basic

HOBBIES

Surfing, reading, graphic design.

PROFESSIONAL EXPERIENCE

FARASHA SYSTEMS | Rabat, Morocco

Electronics Team Manager — Mar 2026 - Present

- Lead avionics architecture, component selection and cross-functional integration. Coordinate validation, field troubleshooting and technical delivery across electronics, embedded, RF and mechanical teams.

UAV & Avionics Systems Engineer — Aug 2025 - Mar 2026

- Integrated aerial and maritime platforms, covering autopilots, sensors, payloads and power systems from simulation and bench checks through flight and water trials.

UAV & Avionics Engineering Consultant — 2023 - Present

Upwork, Fiverr and direct clients

- Provide engineering consultations, system integration, autopilot configuration and tuning. Guide field teams through live calls, log-based troubleshooting, corrective actions and mission preparation.

CAPGEMINI ENGINEERING | Casablanca

Technical Documentation Technician / Consultant — Jul - Sep 2023

- Produced Stellantis after-sales diagnostic and service documentation, following a four-month internship studying automatic emergency braking.

ENGINEERING PROJECTS

Custom Avionics & Companion Computing
Altium Designer / KiCad ▪ H7 / Pixhawk ▪ CM5 / Orin Nano

- Designed flight-controller, ESC and power PCBs, then tested and integrated fabricated boards into real aircraft. Combined autopilot and companion-computer architectures with sensor and vision interfaces, from board bring-up to system validation.

VTOL & QuadPlane Systems
4+1 ▪ Tiltrotor ▪ Tricopter VTOL ▪ Simulation-to-flight validation

- Developed different VTOL configurations, adapting airframes, lift/cruise propulsion, electronics and flight-control setup to each platform. Calibrated and tuned hover, transitions and mission performance through simulation and real flight tests.

Drone-to-Rover Integration
ROS2 ▪ MAVLink ▪ Autonomous docking

- Developed and tested a complete drone-rover system from scratch, integrating mechanics, electronics, companion computing and communications. Validated coordinated control, docking, actuators and failsafes for autonomous operation.

Unmanned Surface Vehicles
ArduRover ▪ Waterjet control ▪ GNSS / RTK

- Delivered a twin-waterjet USV from design and assembly to water trials, addressing stability, differential thrust and autonomous navigation. Integrated power, safety, camera and AI-detection systems; supported AIS on related vessel projects.

Multirotor & Single-Rotor Helicopters
FPV / heavy-lift ▪ Betaflight / INAV ▪ PID / FFT

- Built and tuned multirotors from 2 to 42 inches, matching propulsion, power and avionics for different missions. Supported GPS-denied UAV work and the DARPA Lift Challenge; provided single-rotor helicopter configuration and flight-log diagnostics.

Fixed-Wing UAV Development
Airframe design ▪ Propulsion sizing ▪ ArduPlane / PX4

- Developed V-tail, pusher and twin-motor aircraft using 3D-printed, foam and CNC-cut wood parts. Applied fibreglass/resin reinforcement to printed airframes and flight-tested them. Built custom-model Gazebo/SITL workflows to validate tracking, following and autonomous missions before real flights.